

DARABANI, Romania

WMO: 150000

Lat: 48.1950N

Lon: 26.5736E

Elev: 260

StdP: 98.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-13.1	-17.3	0.8	-14.7	-15.0	1.0	-12.6	13.9	-3.6	12.1	-4.1	3.3	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	31.3	19.7	29.6	19.5	28.1	18.9	21.8	27.7	20.8	27.2	20.1	26.0	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.1	24.6	19.0	14.2	22.5	18.1	13.5	21.9	64.8	27.1	61.3	27.7	58.5	26.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.9	8.5	7.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.9	0.2	3.9	9.9	16.0	19.8	20.8	21.4	16.3	9.6	3.9	-1.1
	DBStd	9.77	5.05	5.29	5.04	4.97	3.81	3.27	2.99	3.51	4.58	4.38	4.76	5.30
	HDD10.0	1571	429	276	194	64	5	0	0	0	4	61	191	345
	HDD18.3	3432	688	509	448	253	89	23	11	12	93	271	433	603
	CDD10.0	1488	0	0	5	62	192	293	333	353	192	49	8	0
	CDD18.3	307	0	0	0	1	17	65	86	106	31	1	0	0
	CDH23.3	2417	0	0	0	9	133	453	629	929	259	3	0	0
	CDH26.7	696	0	0	0	0	18	96	177	341	65	0	0	0
Wind	WSAvg	3.4	3.6	3.8	3.8	4.1	3.5	3.1	2.8	2.9	3.2	3.2	3.6	3.4
Precipitation	PrecAvg	744	46	48	48	52	76	97	100	72	61	48	45	50
	PrecMax	1011	76	95	109	139	158	202	209	193	139	150	85	118
	PrecMin	524	14	22	16	13	33	41	40	22	12	3	1	13
	PrecStd	110	17	17	23	24	33	38	49	39	36	29	18	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	13.7	19.2	24.6	28.7	30.5	32.5	34.2	31.7	23.4	17.1	11.9
		MCWB	5.4	9.1	11.6	14.6	17.5	21.3	20.5	20.3	20.2	15.9	11.6	8.2
	2%	DB	5.6	10.7	15.3	22.5	26.5	29.0	30.5	32.0	28.2	20.7	14.3	9.5
		MCWB	4.0	7.3	9.2	13.8	17.1	20.8	20.1	19.3	18.0	14.8	10.4	7.2
	5%	DB	4.0	8.2	12.8	20.2	24.7	27.7	28.6	30.2	26.1	18.5	12.2	7.5
		MCWB	2.7	5.6	7.9	12.8	16.5	19.5	19.9	19.0	17.2	13.6	9.1	5.8
	10%	DB	2.4	6.5	10.5	18.1	22.8	26.1	26.7	28.4	24.1	16.1	10.3	5.2
		MCWB	1.3	4.7	6.6	12.0	15.4	18.8	19.0	18.6	16.7	12.5	8.0	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	9.9	11.9	15.8	19.7	24.3	22.7	21.9	20.6	17.1	12.2	8.9
		MCDB	6.9	13.5	17.9	21.7	25.4	28.1	29.0	27.9	30.4	20.3	15.5	10.7
	2%	WB	4.3	7.6	10.1	14.6	18.5	22.1	21.8	20.9	19.2	15.5	10.7	7.1
		MCDB	5.2	10.2	14.3	20.4	23.6	26.1	28.4	28.4	25.9	19.5	13.1	9.0
	5%	WB	2.7	6.2	8.6	13.2	17.4	20.6	20.9	20.2	18.1	14.3	9.0	5.7
		MCDB	3.6	8.0	12.4	18.1	22.4	25.7	27.1	26.9	23.7	17.6	10.7	7.2
	10%	WB	1.4	4.8	7.2	12.0	16.2	19.5	20.0	19.4	17.2	13.0	7.6	4.1
		MCDB	2.1	6.4	9.9	16.7	21.2	24.2	25.5	25.3	23.0	15.5	9.1	5.0
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.1	7.6	9.5	9.8	9.5	9.7	10.4	9.0	7.4	5.1	4.6
		MCDBR	6.5	9.4	11.8	13.2	12.3	11.4	12.2	12.7	12.6	11.3	8.1	7.7
		MCWBR	5.1	6.5	7.0	6.3	5.2	5.0	4.5	4.3	4.8	5.6	5.3	5.7
	5% WB	MCDBR	6.3	9.0	10.8	12.2	11.1	10.5	11.6	11.1	11.5	9.5	7.6	6.9
		MCWBR	5.1	6.4	6.5	6.2	5.3	5.2	4.8	4.2	4.7	5.1	5.3	5.5
Clear-Sky Solar Irradiance	taub	0.313	0.343	0.380	0.436	0.409	0.408	0.437	0.442	0.400	0.366	0.340	0.311	
	taud	2.277	2.232	2.228	2.113	2.264	2.294	2.220	2.200	2.275	2.349	2.358	2.338	
	Ebn at Noon	738	797	827	812	854	856	823	798	797	764	704	696	
	Edh at Noon	76	99	117	145	130	127	135	132	111	88	71	64	
All-Sky Solar Radiation	RadAvg	1.07	1.84	3.07	4.44	5.64	5.96	5.85	5.23	3.65	2.29	1.06	0.80	
	RadStd	0.16	0.29	0.33	0.42	0.46	0.48	0.40	0.37	0.48	0.30	0.10	0.12	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.70	N/A	N/A	N/A	+0.86	N/A	N/A	-147	-211	+105	+38

Nomenclature: See separate page